Three-Phase System

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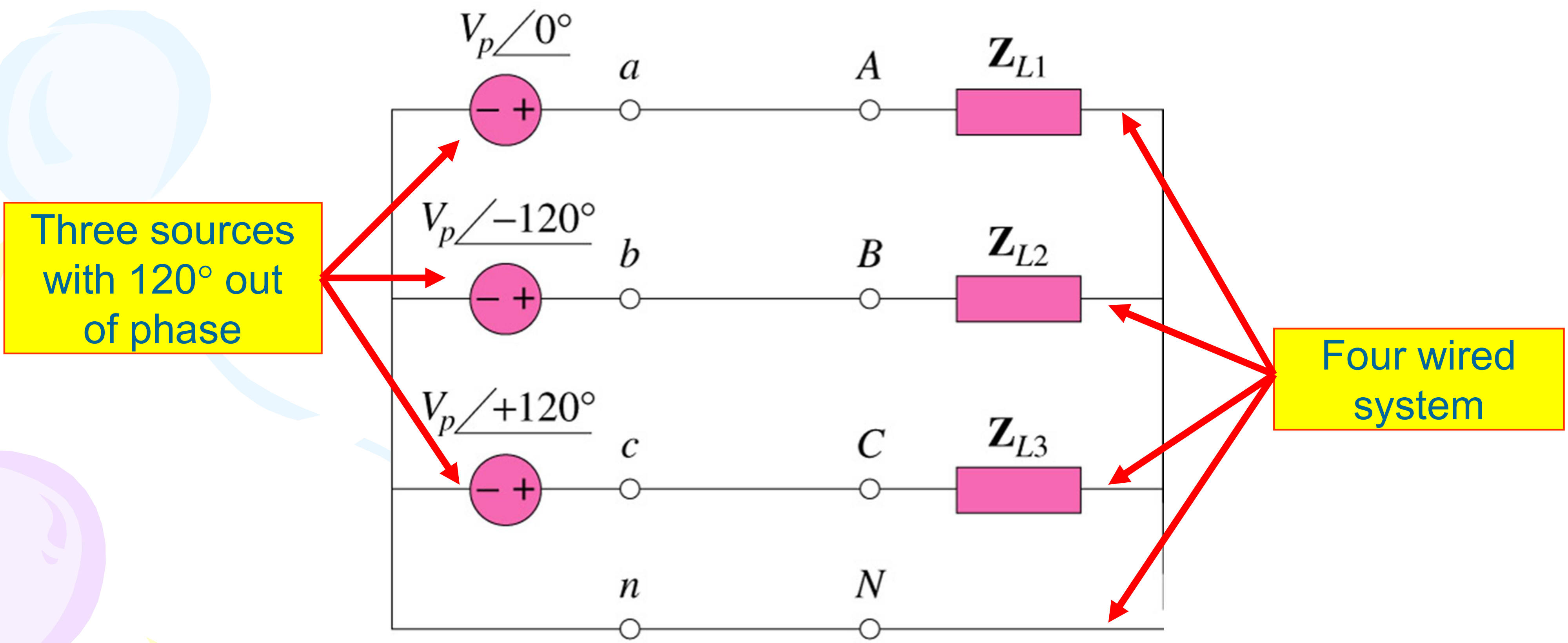
12.1 What is a Three-Phase Circuit?(1)

- It is a system produced by a generator consisting of **three sources** having the same amplitude and frequency but **out of phase** with each other by 120°.



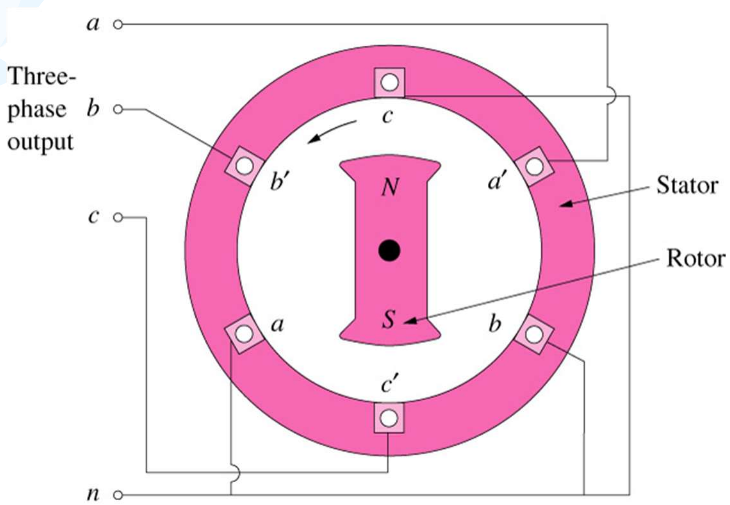
12.1 What is a Three-Phase Circuit?(2)

Advantages:

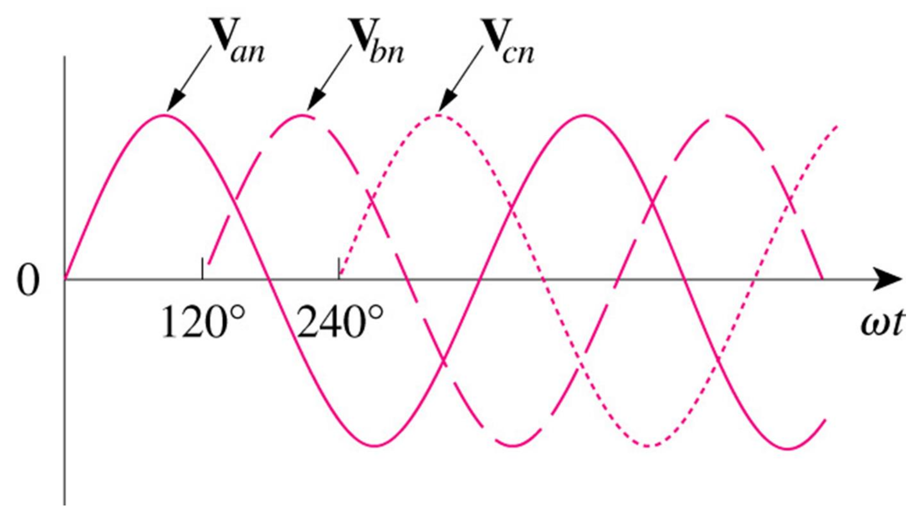
1. Most of the electric power is generated and distributed in three-phase.
2. The instantaneous power in a three-phase system can be constant.
3. The amount of power, the three-phase system is more economical that the single-phase.
4. In fact, the amount of wire required for a three-phase system is less than that required for an equivalent single-phase system.

12.2 Balance Three-Phase Voltages (1)

- A three-phase generator consists of a rotating magnet (rotor) surrounded by a stationary winding (stator).



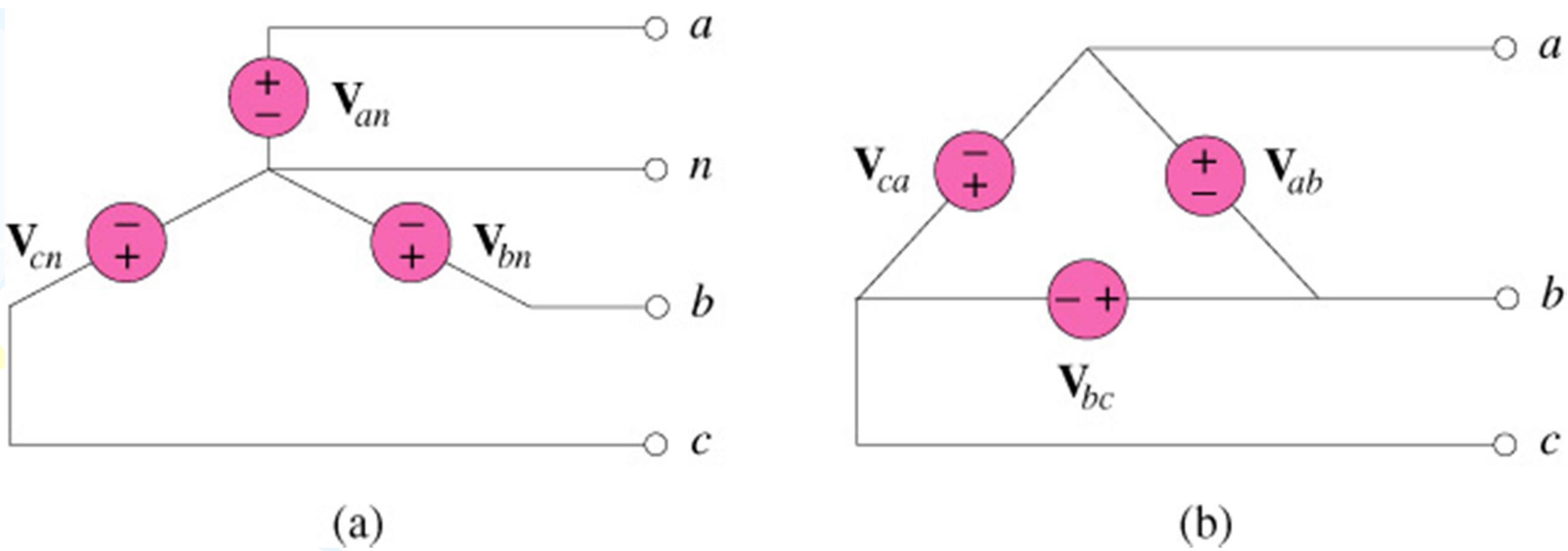
A three-phase generator



The generated voltages

12.2 Balance Three-Phase Voltages (2)

- Two possible configurations:



Three-phase voltage sources: (a) Y-connected ; (b) Δ-connected

12.2 Balance Three-Phase Voltages (3)

- ***Balanced phase voltages*** are equal in magnitude and are out of phase with each other by 120°.
- The ***phase sequence*** is the time order in which the voltages pass through their respective maximum values.
- A ***balanced load*** is one in which the phase impedances are equal in magnitude and in phase

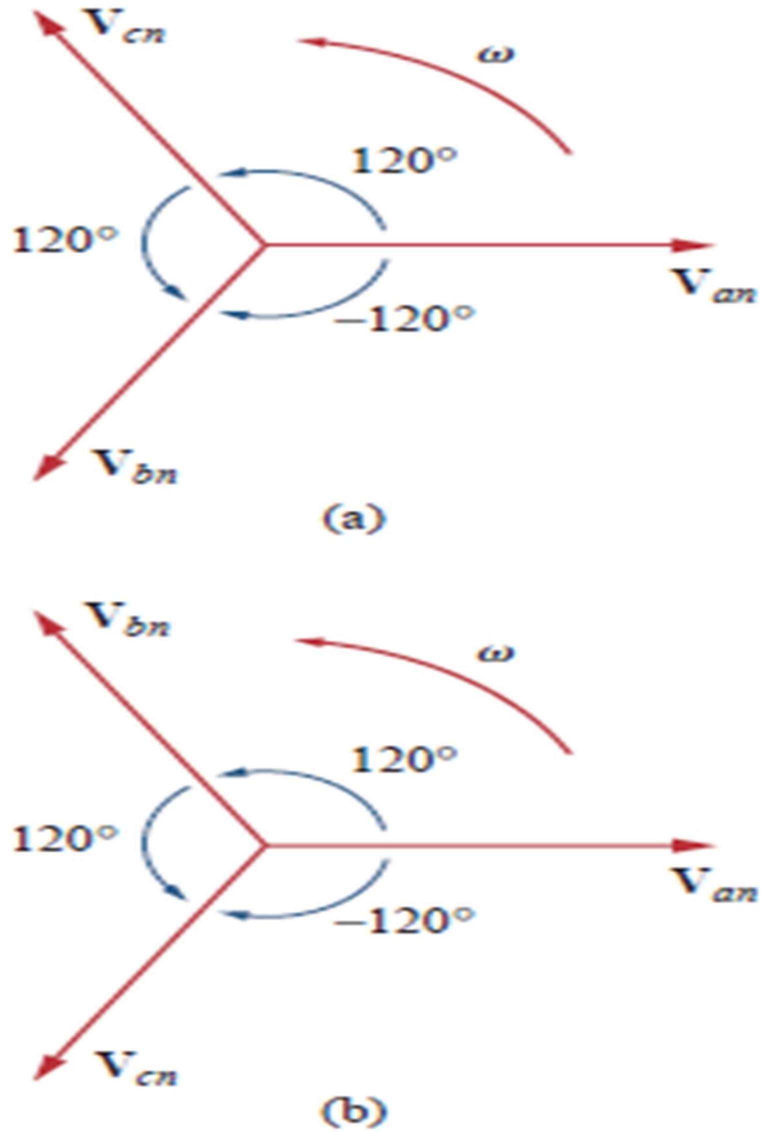


Figure 12.7
Phase sequences: (a) *abc* or positive sequence, (b) *acb* or negative sequence.

$$
\begin{aligned}
\mathbf{V}_{an} &= V_p\angle 0^\circ \\
\mathbf{V}_{cn} &= V_p\angle -120^\circ \\
\mathbf{V}_{bn} &= V_p\angle -240^\circ = V_p\angle +120^\circ
\end{aligned}
\qquad (12.4)
$$

This is called the *acb sequence* or *negative sequence*. For this phase sequence, $\mathbf{V}_{an}$ leads $\mathbf{V}_{cn}$, which in turn leads $\mathbf{V}_{bn}$. The *acb* sequence is produced when the rotor in Fig. 12.4 rotates in the clockwise direction. It is easy to show that the voltages in Eqs. (12.3) or (12.4) satisfy Eqs. (12.1) and (12.2). For example, from Eq. (12.3),

$$
\begin{aligned}
\mathbf{V}_{an} + \mathbf{V}_{bn} + \mathbf{V}_{cn} &= V_p\angle 0^\circ + V_p\angle -120^\circ + V_p\angle +120^\circ \\
&= V_p(1.0 - 0.5 - j0.866 - 0.5 + j0.866) \qquad (12.5) \\
&= 0
\end{aligned}
$$

The **phase sequence** is the time order in which the voltages pass through their respective maximum values.

12.2 Balance Three-Phase Voltages (4)

Example 1

Determine the phase sequence of the set of voltages.

$$v_{an} = 200\cos(\omega t + 10°)$$

$$v_{bn} = 200\cos(\omega t - 230°)$$

$$v_{cn} = 200\cos(\omega t - 110°)$$

12.2 Balance Three-Phase Voltages (5)

Solution:

The voltages can be expressed in phasor form as

$$\mathrm{V}_{an} = 200\angle 10^\circ \text{ V}$$
$$\mathrm{V}_{bn} = 200\angle -230^\circ \text{ V}$$
$$\mathrm{V}_{cn} = 200\angle -110^\circ \text{ V}$$

We notice that $\mathbf{V}_{\mathbf{an}}$ leads $\mathbf{V}_{\mathbf{cn}}$ by 120° and $\mathbf{V}_{\mathbf{cn}}$ in turn leads $\mathbf{V}_{\mathbf{bn}}$ by 120°.

Hence, we have an acb sequence.

Given that $\mathbf{V}_{bn} = 110\angle 30^\circ$ V, find $\mathbf{V}_{an}$ and $\mathbf{V}_{cn}$, assuming a positive (abc) sequence.

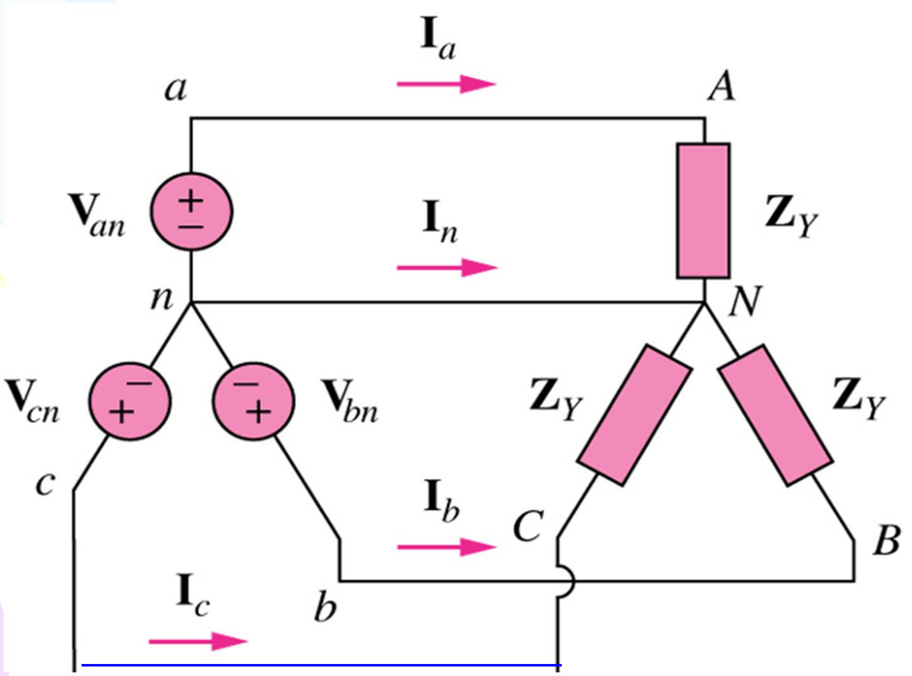
Answer: $110\angle 150^\circ$ V, $110\angle -90^\circ$ V.

12.3 Balance Three-Phase Connection (1)

- Four possible connections
  1. Y-Y connection (Y-connected source with a Y-connected load)
  2. Y-Δ connection (Y-connected source with a Δ-connected load)
  3. Δ-Δ connection
  4. Δ-Y connection

12.3 Balance Three-Phase Connection (2)

- A balanced Y-Y system is a three-phase system with a balanced y-connected source and a balanced y-connected load.

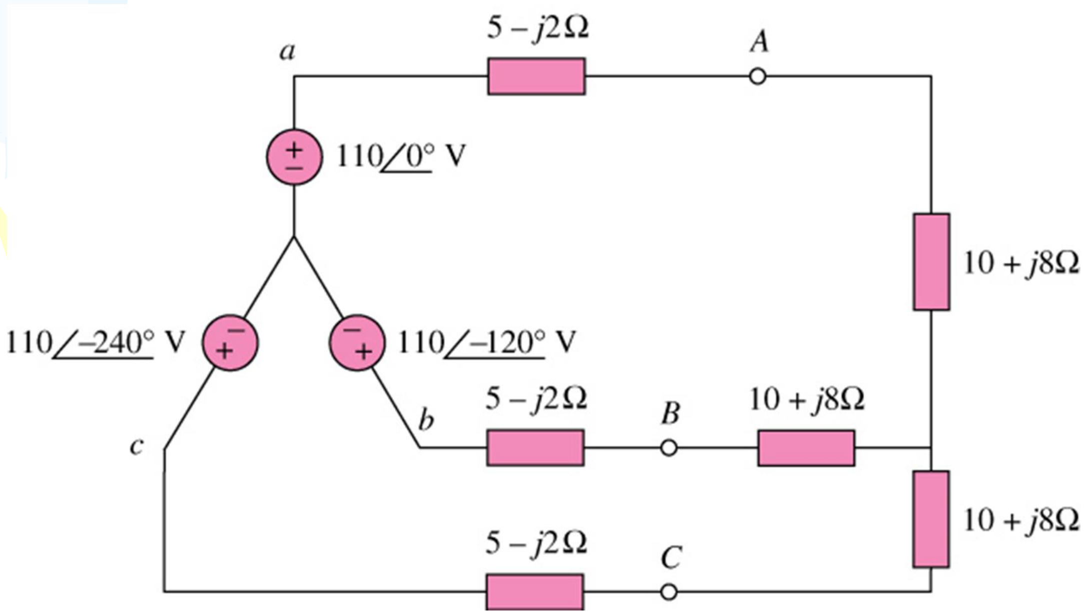


$$V_L = \sqrt{3}V_p \text{ , where}$$
$$V_p = |\mathrm{V}_{an}| = |\mathrm{V}_{bn}| = |\mathrm{V}_{cn}|$$
$$V_L = |\mathrm{V}_{ab}| = |\mathrm{V}_{bc}| = |\mathrm{V}_{ca}|$$

12.3 Balance Three-Phase Connection (3)

Example 2

Calculate the line currents in the three-wire Y-Y system shown below:



Ans

$$I_a = 6.81\angle -21.8^\circ \text{ A}$$

$$I_b = 6.81\angle -141.8^\circ \text{ A}$$

$$I_c = 6.81\angle 98.2^\circ \text{ A}$$

*Refer to in-class illustration, textbook

The three-phase circuit in Fig. 12.13 is balanced; we may replace it with its single-phase equivalent circuit such as in Fig. 12.12. We obtain $\mathbf{I}_a$ from the single-phase analysis as

$$\mathbf{I}_a = \frac{\mathbf{V}_{an}}{\mathbf{Z}_Y}$$

where $\mathbf{Z}_Y = (5 - j2) + (10 + j8) = 15 + j6 = 16.155\underline{/21.8^\circ}$. Hence,

$$\mathbf{I}_a = \frac{110\underline{/0^\circ}}{16.155\underline{/21.8^\circ}} = 6.81\underline{/-21.8^\circ}\text{ A}$$

Since the source voltages in Fig. 12.13 are in positive sequence, the line currents are also in positive sequence:

$$\mathbf{I}_b = \mathbf{I}_a\underline{/-120^\circ} = 6.81\underline{/-141.8^\circ}\text{ A}$$

$$\mathbf{I}_c = \mathbf{I}_a\underline{/-240^\circ} = 6.81\underline{/-261.8^\circ}\text{ A} = 6.81\underline{/98.2^\circ}\text{ A}$$

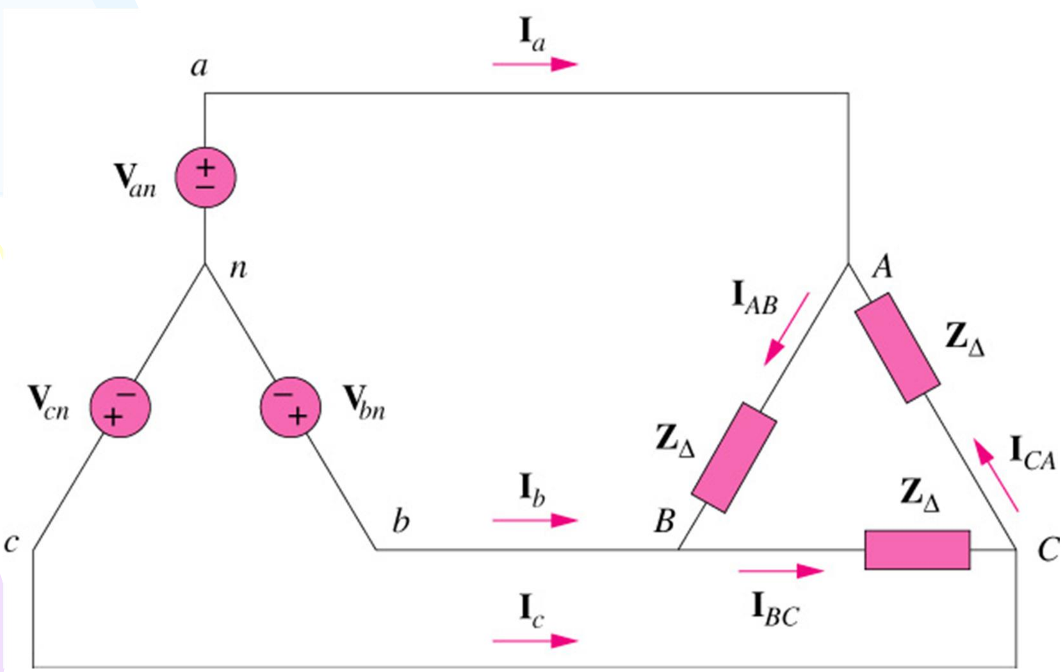
12.3 Balance Three-Phase Connection (4)

- A balanced Y-Δ system is a three-phase system with a balanced y-connected source and a balanced Δ-connected load.

$$I_L = \sqrt{3} I_p \text{ , where}$$

$$I_L = |\mathrm{I}_a| = |\mathrm{I}_b| = |\mathrm{I}_c|$$

$$I_p = |\mathrm{I}_{AB}| = |\mathrm{I}_{BC}| = |\mathrm{I}_{CA}|$$



12.3 Balance Three-Phase Connection (5)

Example 3

A balanced *abc*-sequence Y-connected source with ($V_{an} = 100\angle 10^\circ$) is connected to a Δ-connected load (8+j4)Ω per phase. Calculate the phase and line currents.

Solution

Using single-phase analysis,

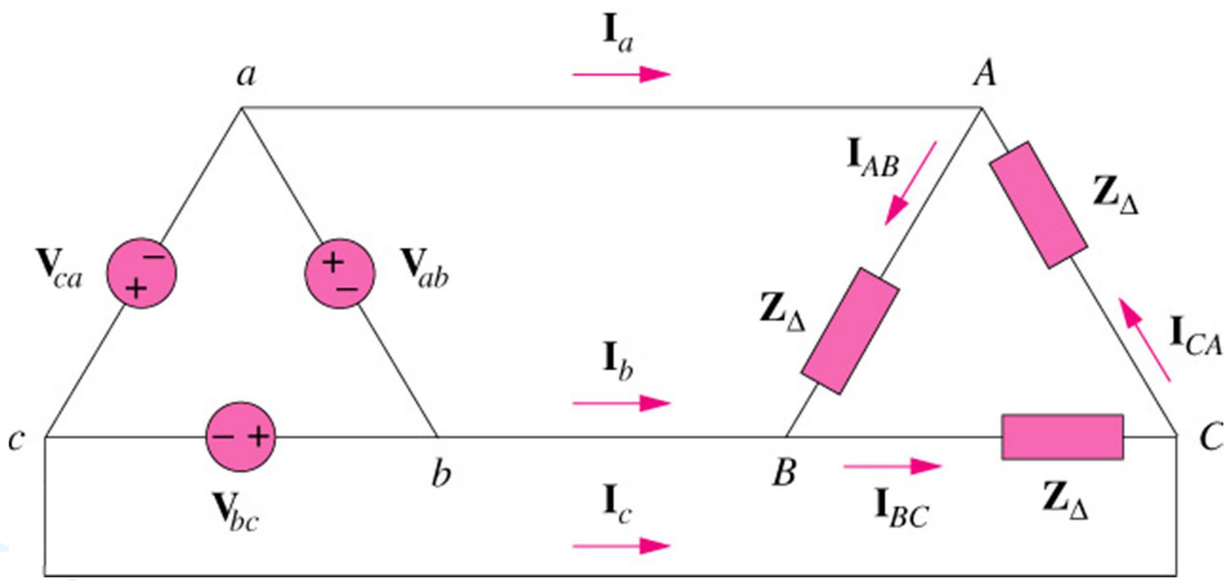
$$I_a = \frac{V_{an}}{Z_\Delta / 3} = \frac{100\angle 10^\circ}{2.981\angle 26.57^\circ} = 33.54\angle -16.57^\circ \text{ A}$$

Other line currents are obtained using the *abc* phase sequence

*Refer to in-class illustration, textbook

12.3 Balance Three-Phase Connection (6)

- A balanced Δ-Δ system is a three-phase system with a balanced Δ -connected source and a balanced Δ -connected load.



12.3 Balance Three-Phase Connection (7)

Example 4

A balanced Δ-connected load having an impedance 20-j15 Ω is connected to a Δ-connected positive-sequence generator having ($V_{ab} = 330\angle 0° \text{ V}$). Calculate the phase currents of the load and the line currents.

Ans:

The phase currents

$$I_{AB} = 13.2\angle 36.87° \text{ A}; I_{BC} = 13.2\angle -81.13° \text{ A}; I_{AB} = 13.2\angle 156.87° \text{ A}$$

The line currents

$$I_a = 22.86\angle 6.87° \text{ A}; I_b = 22.86\angle -113.13° \text{ A}; I_c = 22.86\angle 126.87° \text{ A}$$

*Refer to in-class illustration, textbook

12.3 Balance Three-Phase Connection (8)

- A balanced Δ-Y system is a three-phase system with a balanced y-connected source and a balanced y-connected load.